

Neural Inverse Design of Electrodes Enables Uni-Traveling-Carrier Photodiodes with 220 GHz Bandwidth and 2.7 dBm Output Power at 170 GHz

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Abstract *We demonstrate neural inverse-designed uni-traveling-carrier photodiodes achieving a 220 GHz bandwidth and 2.7 dBm output power at 170 GHz. By tailoring the inductive-capacitive resonance of the on-chip electrodes to compensate for intrinsic RC roll-off, this approach simultaneously enhances bandwidth and output power without heterogeneous integration. ©2026 The Author(s)*

Introduction

Photodiodes for terahertz and microwave photonic systems are required to simultaneously provide high output power and ultra-wide bandwidth [1]. Uni-traveling-carrier photodiodes (UTC-PDs) are promising candidates because only electrons contribute to carrier transport, thereby avoiding the slow hole transport that limits conventional p-i-n photodiodes. Through epitaxial optimization and inductive peaking, bandwidths beyond 330 GHz have been reported [2]. However, the corresponding output power in the D-band generally remains limited, while approaches based on heterogeneous integration often increase fabrication complexity and cost. Fundamentally, this bandwidth-power trade-off arises from the coupled effects of the epitaxial structure, junction capacitance, electrode electromagnetic response, and thermal impedance. These strongly interacting factors define a high-dimensional design space, in which conventional parameter sweeps and experience-based optimization become inefficient. Although machine learning has recently emerged as an effective route for inverse design in complex physical systems [3], its application to high-frequency photodiodes remains limited by broadband performance targets, sparse training data, and the need to optimize multiple device metrics simultaneously.

In this paper, we propose and experimentally demonstrate a neural inverse-design strategy for UTC-PD electrodes. A multi-task neural network (MTNN) is trained to learn the structure-response relationship under limited-data conditions and to inversely generate electrode geometries with designed inductive-capacitive (LC) resonance. This resonance compensates for the intrinsic RC roll-off and flattens the device frequency response without requiring heterogeneous integration. Using this approach, we demonstrate a 3- μm -diameter UTC-PD with a 220 GHz bandwidth and a 14 mA saturated photocurrent, as well as a 6- μm -

diameter UTC-PD delivering 2.7 dBm output power at 170 GHz.

Device Design and Neural Inverse-Design Framework

Figure 1(a) shows the three-dimensional schematic of the proposed device. Two key structural modifications are introduced. First, a ring-shaped n-electrode is adopted instead of conventional point-contact or stripe geometries. By increasing the metal-semiconductor contact perimeter, this design reduces the total series resistance R_s and redistributes the electric field near the active region, thereby improving lateral carrier transport and enhancing the saturation-current capability. Second, to improve optical coupling efficiency, the InP substrate is thinned to 80 μm , and a 264 nm SiO_2 antireflection coating is deposited on the backside. As shown in Fig. 1(b), this modification increases the optical transmittance at 1550 nm from 0.58 to 0.82, corresponding to a 41% relative enhancement.

To design electrodes with the desired LC response, we developed an MTNN tailored for data-limited inverse design. As illustrated in Fig. 1(c), the MTNN consists of shared feature-extraction layers followed by multiple task-specific decoder branches. This architecture enables the model to capture common physical correlations while allowing each decoder to specialize in different target quantities. We first validated the MTNN using coplanar waveguide (CPW) structures, which serve as reliable microwave test vehicles for broadband electrode learning. A dataset containing 207,872 randomly generated CPW geometries was constructed, and full-wave electromagnetic simulations over 0–100 GHz were used to extract the S-parameters, inductance, and Q-factor. The MTNN was trained both as a forward model to predict performance from

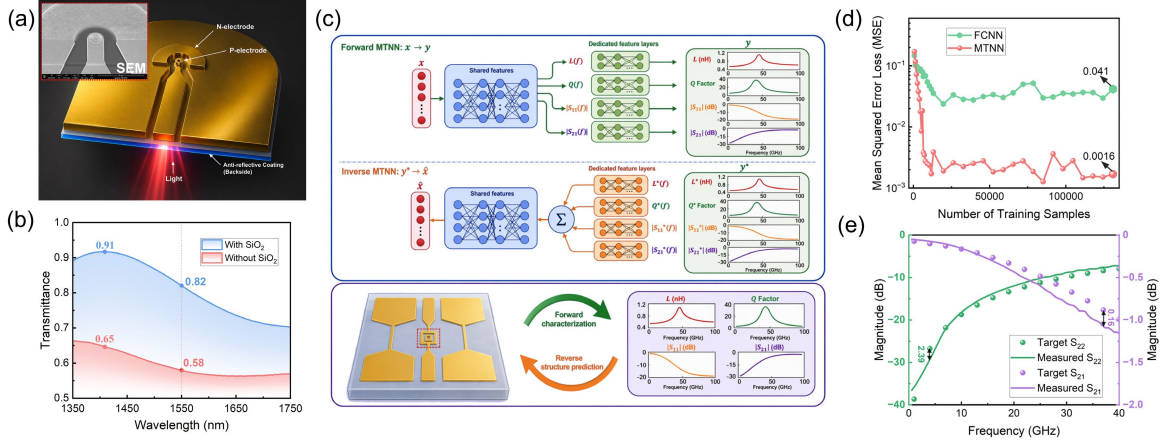


Fig. 1: (a) Three-dimensional schematic and scanning electron microscopy image of the UTC-PD. (b) Optical transmittance with and without backside antireflection coating. (c) Architecture of the proposed MTNN. (d) Validation mean squared error of the FCNN and MTNN. (e) Comparison between target and measured S-parameter characteristic points.

geometry and as an inverse model to predict geometry from prescribed performance targets.

As shown in Fig. 1(d), the MTNN reduces the validation mean squared error by 25.6× compared with a conventional fully connected neural network (FCNN) under the same limited-data conditions. The coefficient of determination R^2 for inverse geometric prediction reaches 0.998, indicating excellent predictive accuracy even in sparse regions of the parameter space. Importantly, the MTNN enables inverse design from sparse frequency-domain specifications, requiring only a limited number of target points at selected key frequencies. Experimental validation using fabricated CPW structures further confirms the effectiveness of the model. The measured S-parameters agree with the target responses within 0.16 dB for insertion loss S_{21} and 2.39 dB for return loss S_{22} over the DC–40 GHz range. The equivalent series inductance at 37 GHz is measured to be 0.36 nH, compared with a target value of 0.30 nH, which remains acceptable considering the strong phase sensitivity near resonance.

Building on the validated MTNN framework, we extended the method to the inverse design of single-port UTC-PD electrodes. As shown in Fig. 2(a), a lumped-element equivalent circuit was established to model the intrinsic device behavior. The circuit includes carrier transit-time effects (C_t , R_t), junction capacitances (C_a , C_c , C_p), and series resistance R_s . A dataset comprising 9,223 randomly generated single-port electrode geometries was then generated through full-wave electromagnetic simulations and used for UTC-PD-specific optimization. The inverse model achieves an R^2 value of 0.9993, indicating highly accurate response-to-geometry prediction.

Our design strategy tailors the electrode inductance to form an LC resonance with the total device capacitance at a target frequency, thereby compensating for the intrinsic RC roll-off and flattening the overall frequency response. Circuit co-simulations were carried out by combining the electrode S-parameter model with the intrinsic equivalent circuit. Characteristic inductance response points centered at 90, 105, 135, and 155 GHz were selected as inverse-design targets.

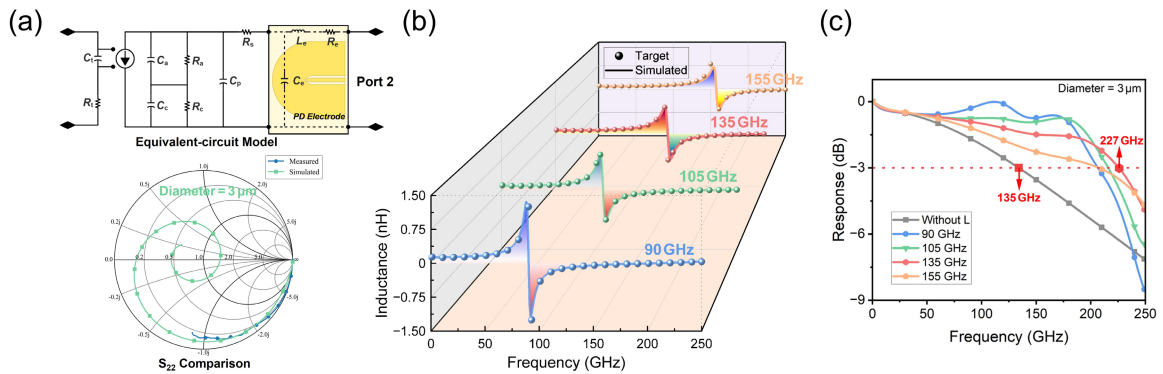


Fig. 2: (a) UTC-PD equivalent circuit and corresponding Smith chart. Yellow region: MTNN inverse-designed electrode modeled as a broadband two-port S-parameter network. (b) Target vs. simulated inductance response of MTNN inverse-designed electrodes. (c) Device frequency responses with MTNN inverse-designed electrodes at different resonant frequencies.

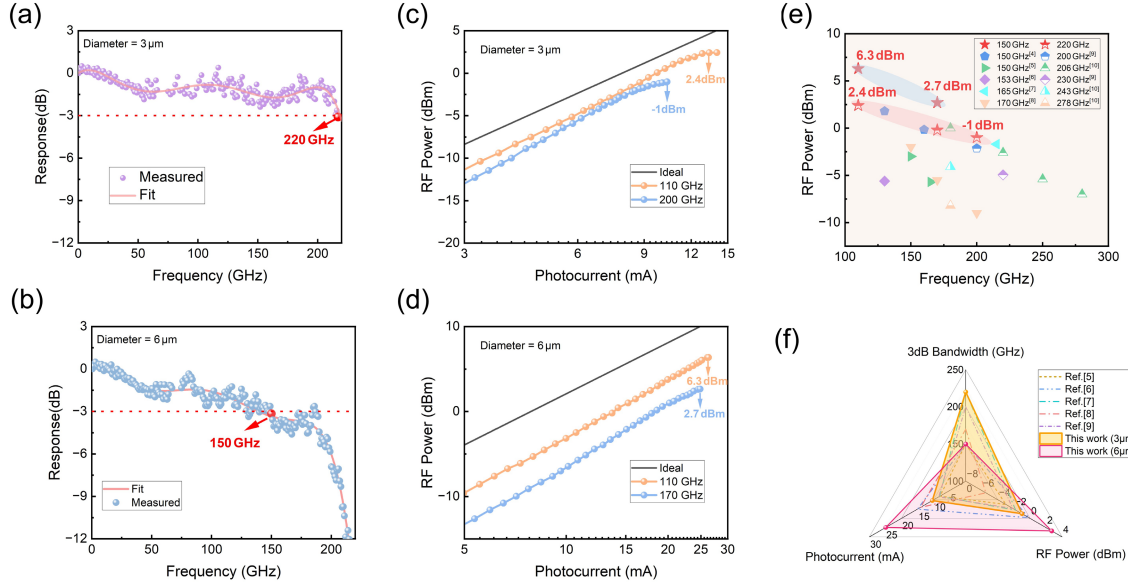


Fig. 3: (a), (b) Measured frequency responses of the 3 μm and 6 μm devices. (c), (d) Saturated output powers of the 3 μm and 6 μm devices at selected frequencies. (e), (f) Performance comparison with previously reported UTC-PDs.

The corresponding electrode geometries predicted by the MTNN were then re-simulated and validated in Ansys HFSS. As shown in Fig. 2(b), the simulated inductance curves agree well with the target characteristic points, confirming that the MTNN can accurately generate electrode geometries for prescribed performance targets. As illustrated in Fig. 2(c), the electrode designed for a resonant frequency of 135 GHz yields a flat response up to 220 GHz and extends the simulated -3 dB bandwidth from 135 GHz to 227 GHz.

Experimental Results

Based on the inverse-designed electrode geometries, UTC-PDs with diameters of 3 μm and 6 μm were fabricated. Under a bias of -2 V, the dark currents were measured to be 1.4 nA and 2.2 nA, respectively, indicating low leakage current and good material quality. Owing to substrate thinning and backside antireflection coating, the external responsivities at 1550 nm reached 0.15 A/W for the 3 μm device and 0.23 A/W for the 6 μm device.

The frequency responses were characterized using a heterodyne measurement setup consisting of two tunable lasers and an RF power meter. As shown in Fig. 3(a), the 3 μm device exhibits a flat response from 75 GHz to 200 GHz and achieves a -3 dB bandwidth of 220 GHz, in good agreement with the co-simulation results. For the 6 μm device, despite its larger junction capacitance, the MTNN-enabled inverse-designed electrode effectively compensates for the high-frequency roll-off, yielding a -3 dB bandwidth of 150 GHz while maintaining a relatively flat response at high frequencies.

The 3 μm device delivers a saturated output power of 2.4 dBm at 110 GHz and -1 dBm at 200

GHz, corresponding to a saturated photocurrent of 14 mA. The 6 μm device achieves output powers of 6.3 dBm at 110 GHz and 2.7 dBm at 170 GHz. As summarized in Fig. 3(e),(f), these results demonstrate that neural inverse electrode design can simultaneously enhance bandwidth and output power, thereby shifting the conventional bandwidth-power trade-off of UTC-PDs toward a more favorable regime.

Conclusion

We have demonstrated high-power, ultra-wide-band UTC-PDs enabled by neural inverse electrode design. A 3 μm UTC-PD achieves a -3 dB bandwidth of 220 GHz with a saturated photocurrent of 14 mA, while a 6 μm UTC-PD delivers 2.7 dBm output power at 170 GHz. These performance gains are realized through optimized on-chip electrodes that introduce a controlled LC resonance to compensate for the intrinsic RC roll-off without requiring heterogeneous integration. The proposed MTNN framework enables accurate inverse optimization under limited-data conditions and reduces the validation error by a factor of 25.6 compared with a conventional FCNN. These results demonstrate that data-driven electrode engineering offers an effective and scalable route toward next-generation terahertz photonic systems.

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